

AGENEWTECH

H164YP

Smart Module Specification_V1.3



Our aim is to provide customers with timely and comprehensive service. For any assistance, please contact our company headquarters:

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About the Document

History

Version	Data	Author	Description
-	2021-12-14	Guanqibin	Initial
1.0	2022-01-04	Guanqibin	Controlled Copy
1.1	2022-05-31	Yang Yanlian	Correct errors
1.2	2024-09-27	Wu Danyi	Update Information
1.3	2026-03-23	Chen Yun	Change company name

H164YP

Multi-network Smart 4G module

Supports Wi-Fi & BT



H164YP is a multi-network Smart LTE Cat 7 module based on MediaTek's Octa-core (Quad-core ARM Cortex-A73+ Quad-core ARM Cortex-A53) processor launched by Wuxi Sunning Technology Co., Ltd, which is equipped with Android 11.0 operating system. Its powerful performance, neon engine processing, 2D/3D graphics acceleration processing capabilities, fully meet the needs of customers for high-speed and multimedia functions in industrial and consumer applications.

H164YP supports full coverage of various network standards, Wi-Fi 802.11a/b/g/n/ac and BT4.2 LE short-range communication. Not only that, H164YP also has audio and video I/O interfaces and rich GPIO interfaces, but also supports a variety of audio and video codecs, and the main screen supports up to FHD 2400×1080. At the same time, the positioning system of GPS+GLONASS+BDS/Galileo are supported by H164YP makes its positioning function fast and accurate.

H164YP is a chip module with 323 pins, including 152 LCC pins and 171 LGA pins. With a size of only 43.0mm×44.0mm×2.6mm, it can be embedded in various M2M products through pads, making it widely used in smart cash registers, smart POS, tax control machines, security monitoring, vehicle-mounted equipment, high-end information collection equipment, intelligent robots, smart homes, smart hardware, industrial smart handheld equipment, unmanned aerial vehicles, high-end police or law enforcement equipment, smart intercom equipment, smart wearing and vending machines.



Main Advantages

- ✓ Comprehensive coverage of multi-network standard.
- ✓ Support Wi-Fi 802.11a/b/g/n/ac and BT4.2 short-range wireless communication
- ✓ Support GPS+GLONASS+BDS/Galileo satellite positioning
- ✓ Support multiple audio and video codecs
- ✓ The main screen supports up to FHD 2400×1080
- ✓ Support 1080P@30fps video recording and playback
- ✓ Possess audio and video input and output interfaces and rich GPIO interfaces



Quad-core ARM Cortex-A73
+
Quad-core ARM Cortex-A53



IEEE 802.11 a/b/g/n/ac



Support Android 11



GPS+GLONASS+BDS/Galileo



Powerful multimedia



BT4.2

LTE Cat7	H164YP
Region/Operator	China/Europe/Asia
Operating system	Android 9/11/14
Memory	64 GB eMMC + 4 GB LPDDR4X(Default) / 32 GB eMMC + 3 GB LPDDR4X(Optional)
Module size (mm)	43.0× 44.0 ×2.6
Packaging	LCC + LGA
Temperature range	
Operation temperature	Normal operating temperature: -20 °C ~ + 80°C Storage temperature range: -40 °C ~ + 90 °C
Frequency band information	
LTE-FDD	B1/B3/B5/B7/B8/B20
LTE-TDD	B34/B38/B39/B40/B41
WCDMA	B1/B2/B5/ B8
GSM/EDGE	B2/B3/B5/ B8
WLAN	2.4/5GHz, 802.11 a/b/g/n/ac
Bluetooth	4.2
GNSS	GPS + GLONASS + BDS/Galileo
Authentication	
Operator Certification	Undetermined
Compulsory Certification	CE/SRRC*/NAL*/CCC*
Other Certifications	RoHS/REACH
Frequency Band Information	
LTE-FDD (Mbps)	300Mbps (DL)/75Mbps (UL)
LTE-TDD (Mbps)	260Mbps (DL)/90Mbps (UL)
DC-HSPA+ (Mbps)	42Mbps (DL)/11.2Mbps (UL)
WCDMA (Kbps)	384Kbps (DL)/384Kbps (UL)
EDGE (Kbps)	296Kbps (DL)/236.8Kbps (UL)
GPRS (Kbps)	107Kbps (DL)/85.6Kbps (UL)
Outstanding characteristics	
Six-mode communication	LTE: Release 12 UMTS: Release 9
Bandwidth (MHz)	1.4 - 20 MHz Radio frequency bandwidth
Charging chip	●
OTG	●
SIM card detection	●
Dual sim dual standby	●
USB Firmware upgrade	●
OTA Firmware upgrade	●

Remarks: 1. *: In progress

2. ●: Supports

3. Extended operating temperature: If the module works in this temperature range, its RF performance might be worse but it can still work properly.

LTE Cat7	H164YP
Electrical specification	
Operating Voltage	3.4~4.4 V, typical value3.8 V
Consumption (typical)	Undetermined
Interface	
LCD	×2, 4-lane MIPI DSI FHD 2400×1080 + DPI HD
Camera	×3, 4-lane MIPI CSI, up to 32MP
TP	Capacitive touch screen
Audio	×3 Analog inputs: 3 groups of analog microphone inputs ×3 Analog outputs: speaker, AB-class receiver and AB-class stereo earphone
USB	USB3.0(only support device mode), USB 2.0(support OTG)
(U)SIM	×2 (1.8V/2.95V)
UART	×2
SD Card	×1
ADC	×1
I2C	×6
SPI	×5
PWRKEY	Internal pull up to VBAT
GPIO	×29
Antenna	Main antenna, Diversity antenna, GNSS antenna, WIFI/BT antenna